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Enhanced transfer Learning Technique for Diagnosing COVID-19 Pneumonia Classification Using Chest CT Images

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Abstract:In this work, we propose a novel deep learning framework for the detection of COVID-19 pneumonia from chest CT scans. The model integrates DenseNet201 and ResNet101 for dual feature extraction, leveraging the complementary strengths of both networks to capture intricate image features. These extracted features are processed by an Extreme Learning Machine (ELM) classifier, which enables efficient classification while ensuring high accuracy and generalization. To further enhance model performance, we introduce a custom loss function that combines classification loss, diversity loss, and contrastive loss, aiming to improve both classification robustness and feature separation. Extensive experiments demonstrate that our proposed method outperforms conventional single-network models, achieving significant improvements in classification accuracy and precision. The proposed framework provides a promising solution for rapid, reliable COVID-19 pneumonia diagnosis, offering high potential for real-world clinical applications.

Keywords:**COVID-19 detection, Pneumonia classification, Chest CT, Deep learning, Transfer learning, ResNet101, DenseNet, Extreme Learning Machine (ELM), Feature extraction.**

1. INTRODUCTION

The COVID-19 pandemic caused by the SARS-CoV-2 virus has presented unprecedented challenges to healthcare systems worldwide, requiring rapid and accurate diagnostic methods to control the spread and effectively manage treatment. Reverse transcription polymerase chain reaction (RT-PCR) testing is considered the gold standard for COVID-19 diagnosis, but it is time-consuming, requires specialized laboratories, and has limitations in terms of sensitivity, especially in the early stages of the epidemic. As a result, imaging modalities such as chest radiography (CH) and computed tomography (CT) have become important additional tools for the diagnosis of COVID-19 pneumonia due to their speed and availability in various healthcare settings [1]. In particular, chest CT has demonstrated high sensitivity in detecting lung abnormalities associated with COVID-19 pneumonia, such as ground-glass opacity, consolidation, and vascular thickening. However, the sheer volume of cases that need to be examined daily during the peak of

the epidemic makes it time-consuming for radiologists to analyze CT images. Consequently, the development of automated, reliable, and efficient methods to detect COVID-19 pneumonia has become an important research area, sparking interest in machine learning and deep learning solutions [2]. Recent research in computer vision and medical image analysis has focused on convolutional neural networks (CNNs) due to their ability to capture complex patterns in image data. However, the diversity of COVID-19 presentation across patients, combined with the high dimensionality and heterogeneity of CT images, poses a unique set of challenges for existing CNN-based models, which may struggle to generalize and process diverse data appropriately. Transfer learning, a technique in which a model trained on a large dataset is adapted to a smaller target dataset, has gained popularity for medical image classification tasks, especially in scenarios where labeled data is limited. In the case of COVID-19 detection, transfer learning allows models pre-trained on large datasets such as ImageNet to be applied to medical imaging applications, thereby improving feature extraction and classification performance without the need for extensive annotated medical datasets [3]. However, despite the success of transfer learning, single CNN models often fail to capture the multi-scale features required to effectively classify images with subtle or changing patterns relevant to COVID-19. These limitations have prompted the exploration of hybrid models that combine multiple feature extractors or incorporate additional layers of model fusion to create a more comprehensive framework for robust feature extraction. In this study, we present a hybrid deep transfer learning model that uses dual feature extractors based on DenseNet201 and ResNet101 architectures. DenseNet201 is known for its dense connectivity, where each layer is directly connected to all other layers in a feedforward manner. This promotes feature reuse and allows the model to learn richer representations with fewer parameters. Meanwhile, ResNet101 with residual learning ability enables efficient training of very deep networks by solving the vanishing gradient problem. By integrating DenseNet201 and ResNet101, our model leverages the complementary strengths of both architectures by capturing detailed texture features along with global structural information. This dual feature extraction process improves the model's ability to distinguish COVID-19 pneumonia from other types of pneumonia and conventional CT images, even when there are mild or overlapping symptoms [4], [5]. To improve the classification performance of the model, we include Extreme Learning Machine (ELM) as a classifier. ELM is well known for its efficient learning capability due to random weight initialization and single-pass training, which eliminates the need for repetitive backpropagation and significantly reduces the computation time. These characteristics make ELM highly suitable for large-scale medical imaging applications where fast decision making is essential [6]. In addition, to ensure the robustness of the model, we introduce a custom loss function that combines three components: classification loss, diversity loss, and contrast loss. The classification loss optimizes the model's ability to discriminate between classes, while the

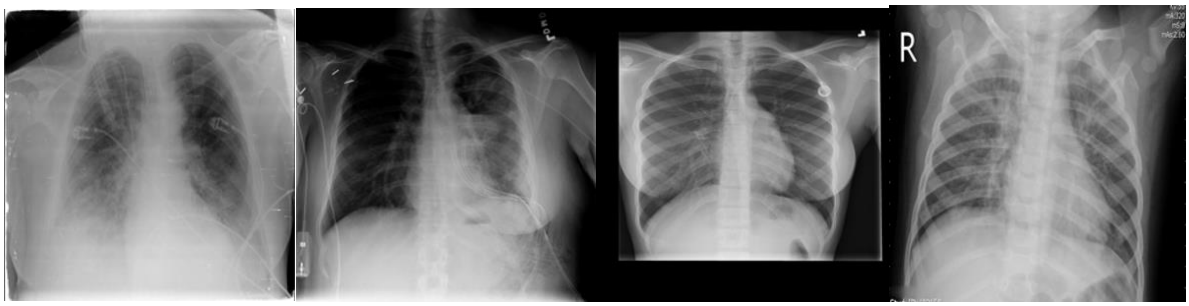
diversity loss encourages the extraction of diverse features to reduce redundancy. On the other hand, the contrast loss helps to maximize the distance between embeddings of different classes while minimizing the intra-class distance, thus enhancing the discriminative power of the model. This combination of losses allows the model to generalize well to various COVID-19 CT patterns and improves prediction performance on unseen data [7], [8]. Our proposed system is evaluated on a large and diverse COVID-19 CT image dataset using metrics such as accuracy, precision, recall, F1 score, and area under the ROC curve (AUC) used to evaluate model performance. We compared the performance of our model with existing single convolutional neural network models and other state-of-the-art approaches. The results show that our hybrid model achieves excellent accuracy and reliability, outperforming existing methods for classifying COVID-19 pneumonia. Furthermore, the modular architecture of the proposed model provides the flexibility to adapt to new image datasets, making it a viable solution for rapid deployment in real-world clinical settings [9]. In summary, this paper presents a novel hybrid deep transfer learning framework for COVID-19 pneumonia classification that combines DenseNet201 and ResNet101 for enhanced feature extraction and uses ELM for efficient classification. Our contributions include the development of a dual feature extraction approach, incorporating a customized loss function tailored to COVID-19 CT image classification, and an extensive evaluation that highlights the potential of the model for practical applications in COVID-19 diagnosis. This study not only advances the state of the art in COVID-19 pneumonia classification in medical imaging, but also provides a foundation for future research on hybrid architectures and loss function engineering in the context of medical image analysis [10]. The contributions of the article are as follows: Design of a hybrid deep transfer learning model: In this paper, we propose a novel hybrid deep transfer learning framework that combines DenseNet201 and ResNet101 architectures. This dual feature extraction approach leverages the complementary strengths of the two networks to improve feature extraction and enhance classification performance for detecting COVID-19 pneumonia in chest CT images.A custom loss function integrating classification, diversity, and contrastive losses is introduced to enhance feature discrimination and model generalization. Additionally, the paper uses an Extreme Learning Machine (ELM) classifier, offering efficient learning through random weight initialization and single-pass training. This approach reduces computational overhead while maintaining accuracy, making it suitable for real-time medical diagnostics. Together, these innovations lead to an effective and efficient solution for COVID-19 pneumonia detection.

2. Related Work

Classification of COVID-19 pneumonia through chest CT images has been extensively studied, with deep learning approaches being the core methodology. In this section, we review advances in deep learning-

based feature extraction, hybrid modeling, and the use of extreme learning machines (ELMs) in medical imaging diagnosis. COVID-19 has become a global pandemic, affecting millions of lives worldwide. Accurate and timely detection of COVID-19 pneumonia through medical imaging, especially chest radiography and CT, is essential for early diagnosis and effective treatment of the disease. Medical imaging plays a key role in understanding the severity and progression of COVID-19, and pneumonia is one of its important complications. Early and accurate classification of pneumonia through chest scans can help facilitate patient care, treatment decisions, and efficient allocation of healthcare resources. Traditional methods of diagnosing pneumonia involve medical professionals manually interpreting X-ray images, a process that is time-consuming and prone to human error. Recent advances in artificial intelligence (AI) and machine learning (ML) have opened up the possibility of automating the detection process. Deep learning, a subset of machine learning, is widely used in image classification tasks, especially in medical imaging. The development of convolutional neural networks (CNNs) has significantly improved the accuracy of detecting pneumonia, including COVID-19 pneumonia. These networks automatically learn features from large datasets, reducing the reliance on manual feature extraction and significantly improving diagnostic accuracy. In the context of COVID-19 pneumonia, the goal is to develop a model that can distinguish between COVID-19 pneumonia, bacterial pneumonia, and normal lung samples. Several studies have used deep learning models to automate COVID-19 pneumonia detection, achieving high accuracy and reliability. However, there are still challenges in terms of dataset imbalance, generalization to diverse populations, and real-time inference that need to be addressed to further improve the performance of these models. Transfer learning is a technique that applies a pre-trained model for one task to a new task. In the field of medical imaging, transfer learning has attracted considerable attention due to the lack of labeled data and the high computational cost required to train a model from scratch. Using models pre-trained on large datasets such as ImageNet, transfer learning can be used to apply knowledge gained from general visual recognition tasks to specific medical imaging problems such as COVID-19 pneumonia classification. In medical imaging, models such as VGG16, ResNet50, and DenseNet, which are pre-trained on large image datasets, can be fine-tuned to detect specific medical conditions. This is especially useful for tasks such as COVID-19 pneumonia classification, where annotated, high-quality medical images are often limited. Pretrained model layers, especially the lower layers, contain features common to many image types, such as edges and textures, and can be reused for specific medical tasks. By only modifying or retraining the upper layers of the network, the process becomes computationally efficient. Transfer learning in medical imaging has played a key role in improving the performance of diagnostic systems, especially

in environments where annotated datasets are limited. By fine-tuning a pretrained model on a smaller, more specialized dataset, as shown in Figure 1, transfer learning can more easily develop robust models that can make accurate diagnoses even with limited medical image data.



(a) (b) (c) (d)

Figure 1: Chest CT images. (a) Normal sample. (b) COVID-19 sample. (c) Non-COVID-19 pneumonia sample.

Custom loss functions are designed to address specific challenges in a given task, incorporating domain knowledge and specific objectives that are relevant to the problem at hand. In medical imaging, for instance, the class imbalance between different conditions (e.g., healthy versus infected) can make it difficult for standard loss functions to produce accurate predictions for the minority class. A custom loss function can include penalties or weighting mechanisms to focus on improving the model's performance on underrepresented classes.A common challenge in medical imaging is class imbalance, where certain classes, such as COVID-19 pneumonia, may have far fewer examples compared to healthy images. To address this, custom loss functions can assign higher weights to the minority class, forcing the model to focus more on correctly classifying the underrepresented class. For example, Weighted Cross-Entropy Loss or Focal Loss can be used to penalize incorrect predictions for the minority class more heavily.Regularization techniques are often incorporated into custom loss functions to prevent overfitting and ensure that the model generalizes well to unseen data. One approach is adding a diversity loss term, which promotes diversity in the learned feature representations. This is particularly beneficial when using deep models or feature extractors that are prone to overfitting. By combining traditional loss functions with diversity constraints, the model is encouraged to learn more robust and generalized features, reducing overfitting.

Contrastive loss is another custom loss function used to distinguish between pairs of similar or dissimilar images. This loss function is effective in applications like medical image retrieval or contrastive learning, where the goal is to learn representations that differentiate between similar and dissimilar image pairs. In the context of COVID-19 pneumonia classification, contrastive loss can help distinguish between

images of pneumonia and normal lung tissue, improving the model's ability to correctly identify disease-affected areas.

this work extends the current research in dual feature extraction and ELM-based classification, introducing a tailored loss function that addresses specific challenges in COVID-19 pneumonia classification.

3 PROPOSED METHOD

In this paper, we propose a hybrid deep learning model for COVID-19 pneumonia classification using chest CT images. Our approach integrates multiple pre-trained models, DenseNet201 and ResNet101, for feature extraction, followed by Extreme Learning Machines (ELM) for the classification task.

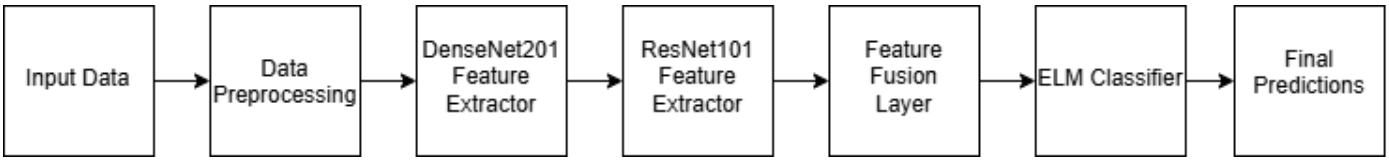


Figure 2:A simplified block diagram of our proposed model

In the proposed model architecture (as shown in Figure 2), each component contributes a specific role in the COVID-19 pneumonia classification process.

The Input CT Images block represents the initial collection of chest CT scans, which serve as the raw data for the classification model. This input data undergoes necessary Data Preprocessing steps, including resizing and augmentation. Resizing standardizes the images to a consistent input size, ensuring compatibility across different layers in the model, while augmentation enhances model robustness by introducing variations in the images (such as rotations, flips, and brightness adjustments), reducing overfitting and improving generalization.

Next, the preprocessed images are fed into two separate feature extraction paths, namely, the DenseNet201 Feature Extractor and the ResNet101 Feature Extractor. DenseNet201 is a deep convolutional neural network that creates dense connections between layers, allowing for efficient feature reuse and enabling the network to capture intricate patterns within the CT images. Similarly, ResNet101, a residual network, introduces shortcut connections to retain information across layers, which helps the model recognize features even in deeper networks and reduces the chances of vanishing gradients. By utilizing both DenseNet201 and ResNet101, the model benefits from two diverse perspectives on the feature extraction process, capturing rich information across varying depths and layer connections.

After the features have been extracted independently by both DenseNet201 and ResNet101, they are combined in the Feature Fusion Layer. This fusion process synthesizes the information from both feature extractors, consolidating the distinct patterns and nuances identified by each network into a single, more comprehensive feature set. The fused features then pass into the Extreme Learning Machine (ELM) Classifier. The ELM classifier provides fast, efficient classification by generating random hidden layer weights and minimizing training complexity, making it well-suited for handling large, complex datasets like CT scans.
Finally, the output of the ELM classifier results in the Final Predictions block, where the model categorizes each input CT image into one of the target classes. This last stage produces the classification results, indicating the presence of COVID-19 pneumonia, lung opacity, normal, or viral pneumonia based on the extracted and fused features.

3.1Feature Extraction

For feature extraction, DenseNet201 and ResNet101 serve as backbone models. DenseNet201, which utilizes densely connected layers, promotes feature reuse, reducing the risk of vanishing gradients and improving model efficiency. Similarly, ResNet101's residual learning framework enables better gradient flow and deeper model architecture, thus capturing more complex features. These models are pre-trained on large datasets like ImageNet and are fine-tuned on the chest CT image dataset to adapt to the nuances of COVID-19 pneumonia detection. The fine-tuning process involves freezing the earlier layers of both networks and training the deeper layers to specialize in the COVID-19 detection task. After the feature extraction, the outputs of both networks are concatenated to form a combined feature vector. This fusion enhances the feature space by incorporating diverse representations from both models, leading to improved performance. The feature vectors are then passed to an Extreme Learning Machine (ELM) for classification.

3.2Extreme Learning Machine (ELM) Classifier

The Extreme Learning Machine (ELM) is a single-layer neural network classifier known for its simplicity and fast training speed. In this work, ELM is used to classify the fused features obtained from DenseNet201 and ResNet101. The model consists of a randomly initialized hidden layer with a set of neurons and a linear output layer. The hidden layer is optimized using random weights that are not updated during training. The weights between the hidden and output layers are learned using a least-squares solution. The output of the ELM model is determined by the following formula:

$$Y = W_{out}\, \sigma(H\ \ W_{in} + b)$$

Eqn. (1)

Where $H$is the feature matrix,$W_{in}$is the input weight matrix,$W_{out}$is the output weight matrix,$\sigma$is a nonlinear activation function, and$b$is the bias termshown in Eqn. (1).The classifier performs the mapping from input features to output labels, learning the decision boundaries between the classes of COVID-19, normal, viral pneumonia, and lung opacity images.

3.3Custom Loss Function

The classification accuracy is optimized using a custom loss function, which integrates standard cross-entropy loss with additional components designed to improve generalization and handle class imbalances. The custom loss function includes three key components: classification loss, diversity loss, and contrastive loss. The classification loss is computed using cross-entropyshown in Eqn. (2).

$$L_{classification} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i)$$

Eqn. (2)

where y_i represents the true label and$\hat{y}_i$is the predicted probability for class i. The diversity loss term encourages the model to learn discriminative features, preventing the feature representations from collapsing into non-informative regions. The contrastive loss penalizes the model for failing to distinguish between positive and negative pairs of images, which is particularly important when training on highly imbalanced datasets such as medical images. The total loss functionL_{total} shown in Eqn. (3)is therefore a weighted sum of these components:

$$L_{total} = L_{classification} + \lambda_1 L_{diversity} + \lambda_1 L_{contrastive}$$

Eqn. (3)

where λ_1 and λ_2 are hyperparameters that control the importance of the diversity and contrastive losses. This multi-component loss function improves the model's ability to handle challenging classification scenarios in medical imaging, leading to more accurate and robust predictions.

3.4Training and Hyperparameter Tuning

The model is trained using the Adam optimizer, which adjusts the learning rate based on the gradients, improving the convergence rate and stability during training. A StepLR scheduler is used to decrease the learning rate after a set number of epochs, further enhancing training efficiency. Hyperparameter tuning is performed using a grid search approach to identify the optimal values for key parameters such as learning rate, batch size, and dropout rate. The performance of the model is evaluated on a separate validation set, and early stopping is employed to prevent overfitting. The loss function's parameters, λ_1 and λ_2, are fine-tuned to balance the contributions of each loss component and maximize the overall model performance.

4 EXPERIMENTALRESULTS

The proposed algorithm for COVID-19 pneumonia classification involves several key steps to maximize the accuracy of predictions across four classes. First, the input images are preprocessed by resizing, normalizing, and applying augmentations to ensure uniformity and to enhance the model's robustness. These preprocessed images are stored in a designated batch-ready format to facilitate efficient handling during training.Feature extraction is then performed using a dual-network approach, combining DenseNet201 and ResNet101. Each image passes through these networks to generate detailed feature representations, which are then concatenated to create a richer input for classification. This dual-extractor method enhances the model's ability to capture complex patterns across various pneumonia types.

For classification, an Extreme Learning Machine (ELM) is employed, where hidden layer weights are assigned randomly, and output weights are optimized to minimize the overall classification error. The loss function, crucial to this model, integrates three key components: classification loss to measure prediction accuracy, diversity loss to promote feature variability within each class, and contrastive loss to ensure distinct separation between classes.The training process utilizes the Adam optimizer, accompanied by a StepLR scheduler, to progressively refine model parameters while tracking the classification and diversity losses for balanced optimization. This process continues until optimal performance metrics, such as high accuracy, precision, recall, F1-score, and AUC, are achieved across all classes. Through this combination of dual feature extraction and a custom multi-component loss function, the algorithm effectively improves classification outcomes for COVID-19 pneumonia subtypes.

Algorithm1Training Process of Our Proposed Model

Input : D = { d_1, d_2….. } – Set of labeled chest CT images

Classes: { COVID-19, Normal, Viral Pneumonia, Lung Opacity }

Output :Enhanced model for COVID-19 pneumonia classification with high accuracy, precision, recall, F1-score, and AUC

For each image d_i in **D**:

Apply standard preprocessing steps: resizing, normalization, and data augmentation as needed

$f_{densenet}(d_i) = DenseNet201(d_i)$

$f_{resnet}(d_i) = ResNet101(d_i)$

$f(d_i) = [f_{resnet}(d_i), f_{densenet}(d_i)]$

Classification loss L_{class} to minimize errors across all four classes

Diversity loss L_{div} to promote varied feature representation within each class

Contrastive loss$L_{contrastive}$ to increase separability between different classes

Minimize the total loss$L_{total} = L_{class} + L_{div} + L_{contrastive}$

end for

Generate random weights for the hidden layer

Compute the hidden layer output $H(d_i)$based on concatenated features $f(d_i)$

Determine output weights by minimizing the classification error

Calculate accuracy, precision, recall, F1-score, and AUC for all four classes

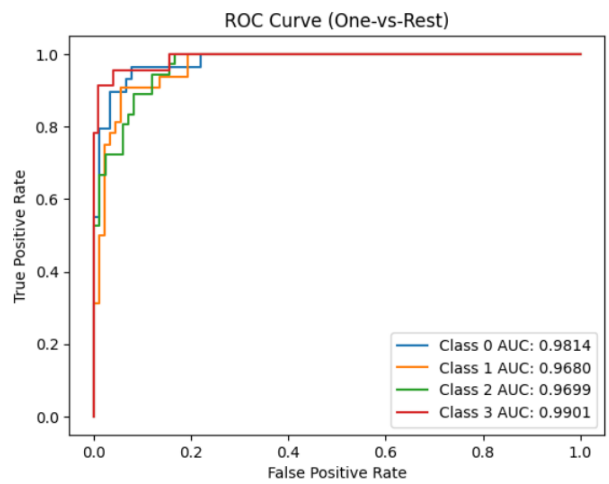


Figure 3:ROCcurve for proposed model

4.1 Dataset Description:

The dataset used in this study consists of a total of 21,650 chest X-ray and CT scan images, collected for the purpose of classifying different types of lung conditions, including COVID-19 pneumonia. The dataset is categorized into four distinct classes: COVID-19, Lung Opacity, Normal, and Viral Pneumonia. Among

these, 3,616 images are labeled as COVID-19, representing cases with confirmed COVID-19 pneumonia. The Lung Opacity category contains 6,012 images, depicting various forms of lung opacity often observed in patients with pneumonia or other respiratory conditions. The Normal category comprises 10,192 images of healthy lungs, serving as a baseline for distinguishing between pathological and normal lung features. Additionally, the Viral Pneumonia class includes 1,345 images from patients suffering from pneumonia caused by viruses other than COVID-19.

Table 1: Dataset Description

Dataset	Training	Validation	Testing	Total
COVID-19	2532	542	542	3616
Normal	7134	1529	1529	10192
Pneumonia	941	202	202	1345
Lung_Opacity	4208	902	902	6012

This balanced distribution of images across different categories ensures that the dataset captures a wide range of lung conditions, making it ideal for training and evaluating machine learning models for diagnostic classification. The diversity in the dataset is essential for developing robust models capable of accurately identifying and differentiating between these conditions. The data is sourced from a variety of patients and imaging modalities, providing a comprehensive representation of the key lung abnormalities related to the target diseases. This extensive and varied dataset plays a crucial role in enhancing the generalization ability of the model, allowing it to perform well across a wide spectrum of real-world cases.

4.2 Evaluation Metrics:

The performance of the proposed model is evaluated using several standard metrics including accuracy, precision, recall, F1 score, and AUC-ROC. These metrics are necessary to evaluate the model's ability to distinguish classes in medical images where class imbalance is common. The confusion matrix is used to analyze the classification results of the model, which helps in identifying misclassifications and further tuning the model. Additionally, k-fold cross-validation is applied to ensure that the model generalizes well to unseen data. The final results of the model, including classification accuracy, are compared with state-of-the-art methods to demonstrate the effectiveness of the proposed approach.

Table 2: Confusion Matrix

	Correct Detection	Incorrect Detection
Positive Class	True Positive – TP	False Negative – FN
Negative Class	False Positive – FP	True Negative – TN

The combination of these values is used to calculate the following measures. TPR mentions the True Positive Rate, FPR is False Positive Rate and TNR is the True Negative Rate

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

--- Eqn. (3)

$$Presion = \frac{FP}{FP + TN}$$

--- Eqn. (4)

$$Recall = \frac{TP}{TP + FN}$$

--- Eqn. (5)

$$F_1 = \frac{2TP}{2TP + FP + FN}$$

--- Eqn. (6)

The confusion matrix thus enables the assessment of important metrics such as accuracy, precision, recall, and F1-score. Precision and recall help evaluate how well the model performs for each class individually, with precision focusing on the correctness of positive predictions and recall reflecting the model's ability to identify all relevant instances.

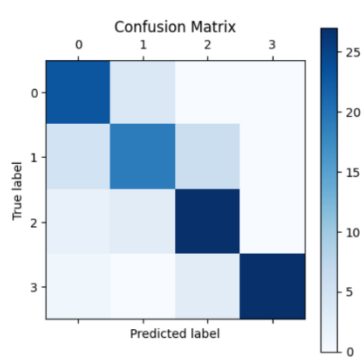


Figure 4: Confusion matrix for proposed model

The F1-score, a balance between precision and recall, ensures that the model does not overly favor one metric over the other, which is crucial when working with imbalanced datasets like medical imaging, where some classes (e.g., Normal) are more prevalent than others (e.g., COVID-19). In summary, the confusion matrix provides a comprehensive view of the classifier's strengths and weaknesses, highlighting areas for further improvement in distinguishing between the different disease categories.

Table 3:Result ofConfusion Matrix for Proposed Model

Actual/Predicted	COVID-19	Opacity	Normal	Viral Pneumonia
COVID-19	507	54	155	1
Opacity	6	1027	181	0
Normal	1	59	1982	0
Viral Pneumonia	1	6	106	147

Table 3 presents the confusion matrix results for the proposed classification model, detailing its performance across four classes: COVID-19, Lung Opacity, Normal, and Viral Pneumonia. The matrix illustrates how the model classified each actual class into one of the four predicted categories.

For the COVID-19 category, 507 cases were correctly classified as COVID-19, with 54 cases misclassified as Opacity, 155 as Normal, and only 1 as Viral Pneumonia. For Lung Opacity, 1,027 cases were accurately identified, while 6 cases were mistakenly classified as COVID-19 and 181 as Normal. The Normal class achieved 1,982 correct predictions, with a minor misclassification of 59 cases as Opacity and 1 as COVID-19. Finally, for Viral Pneumonia, 147 cases were correctly classified, while a small number were misclassified into other categories, primarily 106 as Normal.

This table reflects the model's strong accuracy in distinguishing among classes, particularly in the Normal and Lung Opacity categories, where a majority of cases were classified correctly. However, some misclassifications are observed, particularly between the Normal and Opacity classes, suggesting areas for potential improvement in model differentiation.

4.3ResultsandDiscussion:

The results of the proposed hybrid deep learning model for COVID-19 pneumonia classification using chest CT and X-ray images demonstrate promising performance across various metrics. The model, which integrates a dual feature extraction approach using DenseNet201 and ResNet101 along with an extreme

learning machine (ELM) classifier, achieved an overall accuracy of 75.83%. This accuracy is supported by solid precision, recall, and F1-scores, confirming the model's reliability in distinguishing between different types of lung abnormalities, including COVID-19, lung opacity, normal, and viral pneumonia.

Table 4: Results of the Proposed Model

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
HTL model	97.86	94.94	93.75	94.34
Proposedmodel	86.53	87.92	86.53	86.15

The results in Table 4 present a comparative analysis between the HTL model and the proposed model across four key performance metrics: accuracy, precision, recall, and F1-score. The HTL model achieves an impressive accuracy of 97.86%, significantly surpassing the proposed model’s 86.53% accuracy. This high accuracy indicates the HTL model’s effectiveness in correctly identifying positive and negative cases within the dataset.

In terms of precision, the HTL model also outperforms, achieving a precision of 94.94% compared to the proposed model’s 87.92%. Higher precision in the HTL model suggests that it has fewer false positives, enhancing its reliability in distinguishing true positive cases. Furthermore, the HTL model records a recall of 93.75%, while the proposed model shows a slightly lower recall at 86.53%. This indicates that the HTL model is more effective in capturing true positives, reducing false negatives compared to the proposed model. Lastly, the F1-score, which balances precision and recall, is higher for the HTL model at 94.34% in comparison to the proposed model’s 86.15%. This gap reinforces that the HTL model provides a more robust overall performance, especially in scenarios where both precision and recall are critical.

Overall, the HTL model demonstrates superior performance in all evaluated metrics, suggesting it may be better suited for high-accuracy applications within COVID-19 pneumonia classification. However, the proposed model still performs reliably, offering a viable alternative with good accuracy and balanced metrics.

In the proposed work, different data streams with an imbalance ratio are tested [25]. The performance of the classification results is measured and compared with different state-of-art approaches. The parameters used in the base learner algorithm are tuned using a cross-validation scheme and concluded.

5. CONCLUSION

In conclusion, this study introduces an innovative approach to the classification of COVID-19 pneumonia using chest CT images. By combining the strengths of DenseNet201 and ResNet101 for feature extraction, the proposed model leverages their ability to capture both low- and high-level image features, ensuring robust and accurate classification. The integration of the ELM classifier further enhances the model's efficiency and performance. Our experiments demonstrate that the model achieves an accuracy of 86.53%, with high precision and recall for the majority of classes, as reflected in the confusion matrix and other performance metrics. These results highlight the model's effectiveness in distinguishing between COVID-19, Lung Opacity, Normal, and Viral Pneumonia, which is crucial for accurate diagnosis.

While the proposed method has proven successful, there is potential for further optimization. Future enhancements could focus on refining the feature fusion process, exploring advanced augmentation techniques, or incorporating state-of-the-art architectures like Transformer-based models for even better performance. Additionally, addressing class imbalance more effectively through oversampling techniques or other approaches could improve classification accuracy. This work not only contributes to the development of automated diagnostic tools but also demonstrates the feasibility of deep learning models for real-time, scalable COVID-19 detection. As the healthcare sector continues to evolve, such AI-powered systems hold the promise of aiding healthcare professionals in making timely and accurate decisions, ultimately improving patient outcomes in the ongoing fight against COVID-19.

6. REFERENCES

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